Example 8.3

Number of deaths recorded in various age groups in two areas, *viz.*, Area I and Area II and the population size in each age group are given in the following table.

Age Group (in years)	Area I		Area II	
	Population	No. of Deaths	Population	No. of Deaths
0-10	3000	55	7500	300
10-25	4500	30	6000	50
25-45	6000	40	8000	40
45 and over	1000	15	2000	64

Find the crude death rates and age specific death rates of Area I and Area II.

Solution:

Age Specific Death Rate can be calculated for each age group using the formula

$$ASDR(x,n) = \frac{D(x,n)}{P(x,n)} \times 1000$$

Calculation of the *ASDR* for both the areas in each age group is presented in the following table:

Age Group (x)	Area I			Area II		
	$P(x,n)$	$D(x,n)$	ASDR(x,n) (per thousand)	$P(x,n)$	$D(x,n)$	ASDR(x,n) (per thousand)
0-10	3000	55	$\frac{55}{3000} \times 1000 = 18.33$	7500	300	$\frac{300}{7500} \times 1000 = 40.00$
10-25	4500	30	$\frac{30}{4500} \times 1000 = 6.67$	6000	50	$\frac{50}{6000} \times 1000 = 8.33$
25-45	6000	40	$\frac{40}{6000} \times 1000 = 6.67$	8000	40	$\frac{40}{8000} \times 1000 = 5.00$
45 and over	1000	15	$\frac{15}{1000} \times 1000 = 15.00$	2000	64	$\frac{64}{2000} \times 1000 = 32.00$
Total	14500	140		23500	454	

$$CDR \text{ of Area I} = \frac{140}{14500} \times 1000 = 9.66 \textit{ per thousand}$$

$$CDR \text{ of Area II} = \frac{454}{23500} \times 1000 = 19.32 \textit{ per thousand}$$

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